

Influential Papers on High-Quality Dataset Creation for Large Language Models

This document presents 5 influential research papers focusing on high-quality dataset creation for Large Language Models. Each paper has been cited extensively (at least 50 times) and was published within the last 3 years (2022-2025).

1. Textbooks Are All You Need

<https://arxiv.org/abs/2306.11644>

Abstract:

We introduce phi-1, a new large language model for code, with significantly smaller size than competing models: phi-1 is a Transformer-based model with 1.3B parameters, trained for 4 days on 8 A100s, using a selection of "textbook quality" data from the web (6B tokens) and synthetically generated textbooks and exercises with GPT-3.5 (1B tokens). Despite this small scale, phi-1 attains pass@1 accuracy 50.6% on HumanEval and 55.5% on MBPP. It also displays surprising emergent properties compared to phi-1-base, our model before our finetuning stage on a dataset of coding exercises, and phi-1-small, a smaller model with 350M parameters trained with the same pipeline as phi-1 that still achieves 45% on HumanEval.

2. LIMA: Less Is More for Alignment

<https://arxiv.org/abs/2305.11206>

Abstract:

Large language models are trained in two stages: (1) unsupervised pretraining from raw text, to learn general-purpose representations, and (2) large scale instruction tuning and reinforcement learning, to better align to end tasks and user preferences. We measure the relative importance of these two stages by training LIMA, a 65B parameter LLaMa language model fine-tuned with the standard supervised loss on only 1,000 carefully curated prompts and responses, without any reinforcement learning or human preference modeling. LIMA demonstrates remarkably strong performance, learning to follow specific response formats from only a handful of

examples in the training data, including complex queries that range from planning trip itineraries to speculating about alternate history. Moreover, the model tends to generalize well to unseen tasks that did not appear in the training data. In a controlled human study, responses from LIMA are either equivalent or strictly preferred to GPT-4 in 43% of cases; this statistic is as high as 58% when compared to Bard and 65% versus DaVinci003, which was trained with human feedback. Taken together, these results strongly suggest that almost all knowledge in large language models is learned during pretraining, and only limited instruction tuning data is necessary to teach models to produce high quality output.

3. AlpaGasus: Training A Better Alpaca with Fewer Data

<https://arxiv.org/abs/2307.08701>

Abstract:

Large language models (LLMs) strengthen instruction-following capability through instruction-finetuning (IFT) on supervised instruction/response data. However, widely used IFT datasets (e.g., Alpaca's 52k data) surprisingly contain many low-quality instances with incorrect or irrelevant responses, which are misleading and detrimental to IFT. In this paper, we propose a simple and effective data selection strategy that automatically identifies and filters out low-quality data using a strong LLM (e.g., ChatGPT). To this end, we introduce AlpaGasus, which is finetuned on only 9k high-quality data filtered from the 52k Alpaca data. AlpaGasus significantly outperforms the original Alpaca as evaluated by GPT-4 on multiple test sets and the controlled human evaluation. Its 13B variant matches >90% performance of its teacher LLM (i.e., Text-Davinci-003 generating the 52k data) on test tasks. It also provides 5.7x faster training, reducing the training time for a 7B variant from 80 minutes (for Alpaca) to 14 minutes. Moreover, the experiments prove the efficacy of our method across diverse datasets, base models, and LLM filters. Overall, AlpaGasus demonstrates a novel data-centric IFT paradigm that can be generally applied to instruction-tuning data, leading to faster training and better instruction-following models.

4. What Makes Good Data for Alignment? A Comprehensive Study of Automatic Data Selection in Instruction Tuning

<https://arxiv.org/abs/2312.15685>

Abstract:

Instruction tuning is a standard technique employed to align large language models to end tasks and user preferences after the initial pretraining phase. Recent research indicates the critical role of data engineering in instruction tuning -- when appropriately selected, only limited data is necessary to achieve superior performance. However, we still lack a principled understanding of what makes good instruction tuning data for alignment, and how we should select data automatically and effectively. In this work, we delve deeply into automatic data selection strategies for alignment. We start with controlled studies to measure data across three dimensions: complexity, quality, and diversity, along which we examine existing methods and introduce novel techniques for enhanced data measurement. Subsequently, we propose a simple strategy to select data samples based on the measurement. We present deita (short for Data-Efficient Instruction Tuning for Alignment), a series of models fine-tuned from LLaMA and Mistral models using data samples automatically selected with our proposed approach. Empirically, deita performs better or on par with the state-of-the-art open-source alignment models with only 6K SFT training data samples -- over 10x less than the data used in the baselines. When further trained with direct preference optimization (DPO), deita-Mistral-7B + DPO trained with 6K SFT and 10K DPO samples achieve 7.55 MT-Bench and 90.06% AlpacaEval scores.

5. Datasets for Large Language Models: A Comprehensive Survey

<https://arxiv.org/abs/2402.18041>

Abstract:

This paper embarks on an exploration into the Large Language Model (LLM) datasets, which play a crucial role in the remarkable advancements of LLMs. The datasets serve as the foundational infrastructure analogous to a root system that sustains and nurtures the development of LLMs. Consequently, examination of these datasets emerges as a critical topic in research. In order to address the current lack of a comprehensive overview and thorough analysis of LLM datasets, and to gain insights into their current status and future trends, this survey consolidates and categorizes the fundamental aspects of LLM datasets from five perspectives: (1) Pre-training Corpora; (2) Instruction Fine-tuning Datasets; (3) Preference Datasets; (4) Evaluation Datasets; (5) Traditional Natural Language Processing (NLP) Datasets. The survey sheds light on the prevailing challenges and points out potential avenues for future investigation.

Summary

These five papers collectively represent major advances in understanding and creating high-quality datasets for Large Language Models:

1. **Quality over Quantity:** Papers like "Textbooks Are All You Need" and "LIMA" demonstrate that carefully curated, high-quality data can match or exceed the performance of models trained on orders of magnitude more data.
2. **Automated Quality Assessment:** "AlpaGasus" and "What Makes Good Data for Alignment?" introduce systematic methods for automatically identifying and filtering high-quality training data, making data curation more scalable.
3. **Comprehensive Understanding:** The survey paper provides essential context by cataloging the landscape of available datasets and identifying key challenges and opportunities in dataset creation.

Together, these works establish that the future of LLM development lies not just in scaling up data volume, but in the intelligent curation, filtering, and quality assessment of training data.